4 STRUCTURE CONSIDERATIONS

4.1 INTRODUCTION

Location: Duliajan , Assam

Earthquake Zone: Zone V

The scope of work for this report includes aspects of design methodology considered for structure design/drawings of buildings and other structure in this project. This report generally covers the basis on which structural analysis, design and details will be done. The report covers:

- Structural systems
- Material
- Loading
- Design limit state combinations
- Applicable codes and standards and reference documents.

Data referred/required for design

- Approved layout Plan of the area,
- Approved Architecture drawings,
- Survey Plan of the concerned area,
- GT Report- GT Report should be borehole wise specific to the location where structure has to be constructed.

4.2 DESIGN AND DETAILING METHODOLOGY

Limit State design methods is adopted for the design of the structure. In this method, each member of the structure is designed to satisfy Serviceability criteria, and ultimate collapse criteria. This ensures satisfactory behavior and low probability of failure. The structure system consists of frame structure with conventional beam slab system. The reinforced concrete elements will be designed as per limit state method given in IS 456-2000, Handbook of reinforced concrete detailing- SP34, the steel elements will be designed as per the limit state method given in IS 800:2007.

4.3 STRUCTURAL SYSTEM

The structural system is as following:

- G+6 Storey RCC Building - Ductile RC structural walls with RC SMRFs
- Other RCC buildings-Special Moment Resisting Frame

The project is located in a seismic zone classified as V according to IS1893 (Part1): 2016, the design must consider minimizing the structural damages resulting from an earthquake & basic wind speed classified as 50m/s according to IS 875 part III.

The main considerations followed for the design of structure are:

- Structural safety and stability
- To meet the demands of aesthetics conceived by the architect
- Availability of material, equipment and expertise.
- Constructability and ease of maintenance.
- Durability
- Economy

4.3.1 SUPERSTRUCTURE

The structural system for RCC floors will comprise of RCC column and shear walls framed with RCC Beams & Slab. The stability of building against lateral forces will be entirely provided by the ductile RC structural walls with RC SMRFs for G+6 storey building and by special moment resisting frame (SMRF) for all other buildings in the scheme.

4.3.2 SUBSTRUCTURE

The substructure i.e. the foundation system will consist of pile foundation with pile cap for G+6 storey building and Pile/raft/isolated/combined footings for all other buildings on the recommendations to be made by geotechnical consultant in the GT Report.

4.3.3 RETAINING WALLS

Retaining walls shall be designed for stability (overturning and sliding) and checked for bending & shear. Active earth pressure coefficient shall be applied based on the retained soil fill at the back of walls. To avoid shrinkage cracks on later stage, Shrinkage strips/control joint shall be provided at regular intervals. Foundation of retaining wall structures shall be provided as per soil report. Retaining walls shall be analysed and designed as Beam element. All possible loading of earth pressure, surcharge, fire tender and water pressure shall be applied to retaining wall.

4.3.4 WATER RETAINING STRUCTURES

Water retaining structures shall be designed as per IS: 456-2000 and IS: 3370-2009. Since water retaining structures are subjected to cyclic loading (full or empty conditions) thus as structural

Elements	:	Minimum Concrete Grade
Beams & Slab	:	M30
Columns	:	M30
Retaining Wall	:	M30
Foundation	:	M30
Piles and Pile cap	:	M30
Lean Concrete (PCC)	:	1:4:8/1:3:6 (M15/M10)

4.3.9 REINFORCEMENT AND STRUCTURAL STEEL SPECIFICATION

Grade of reinforcement	: Fe550D for all RCC components
Structural Steel (built-up sections)	: 350MPa
Base plates/connection plates	: 350MPa
Hot-Rolled Standard Sections	: 350MPa
Tubular Sections	: 240MPa / 310MPa
Anchor Bolts	: 4.6, 4.8 & 8.8

4.3.10 EXCAVATION

Excavation shall be done as per structural drawing for excavation. It is contractor's responsibility to design earth retaining system and submit design along with calculation for review and approval by the site in charge. In case of water-logged site, water table shall be brought down by installing discharge point. Contractor shall prepare scheme for discharging water volume from site.

4.4 STRUCTURAL MODELING AND SOFTWARE

(ETABS / STAAD Pro, MS EXCEL, AUTOCAD, etc.)

3-D Analysis of all the buildings Structures will be carried out using ETABS / STAAD Pro Software package and analysis/design results shall be used for designing/detailing the various elements. All the beams and columns will be idealized as Beam Elements. The structure shall be analyzed and designed for all possible Combinations of gravity loads (dead and live load), and lateral loads (Earthquake/Wind loads).

Apart from ETABS/STAAD, design of Foundations, Slab, etc shall be done as per the Standard methods (In-house MS Excel sheet) and relevant codal provisions. Drawings shall be prepared in Auto-cad.

4.5 LOADING PARAMETERS

4.5.1 MATERIAL PROPERTIES

Unit Weight of Reinforced Concrete	=	25.0 kN/m³
Unit Weight of Plain Concrete	=	24.0 kN/ m³
Unit Weight of Steel	=	78.5 kN/ m³
Unit Weight of AAC block	=	8.5 kN/ m³
Modulus of elasticity of steel (Es)	=	2.1 x 105 N/mm²
Modulus of elasticity of concrete (Ec)	=	5000√fck N/mm²
Dry density of Soil	=	18.0 KN/m3
Density of saturated Soil	=	18.0 KN/m3
Density of Water	=	10.0 KN/m3

4.5.2 DEAD LOADS (DL)

Self-Weight

Self-weight of RCC/steel column and beams are automatically considered by the analysis software i.e. (ETABS / STAAD Pro.)

Dead load at RCC floor level shall be as follows:

200mm Slab (0.200 x 25)	=	5.00 KN/m2
150mm Slab (0.150 x 25)	=	3.75 KN/m2
125mm Slab (0.125 x 25)	=	3.125 KN/m2

Wall Brick Load

100mm brick	=	0.85 KN/m/meter height
225mm brick	=	1.9125 KN/m/meter height